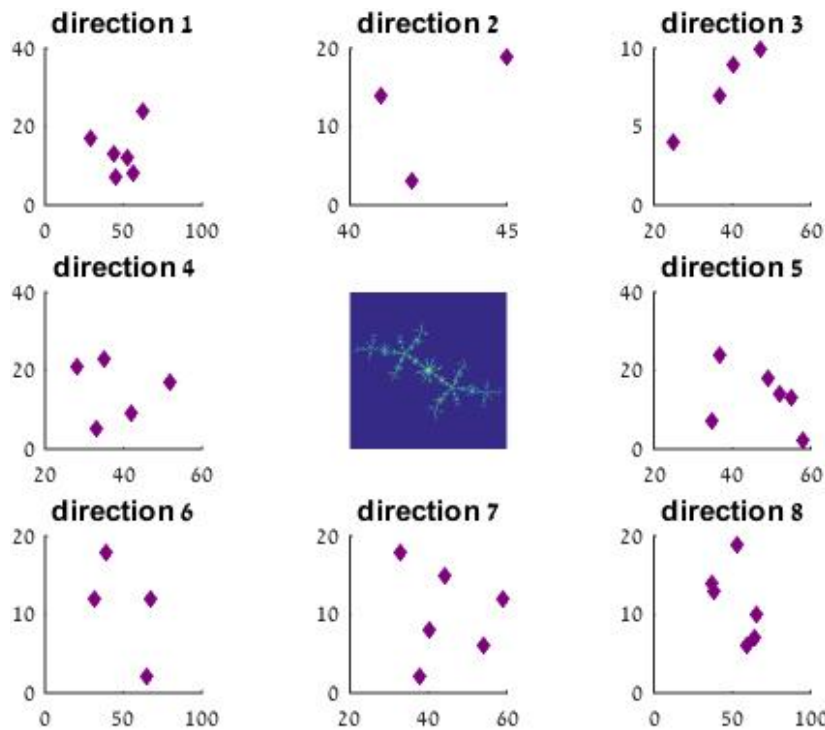




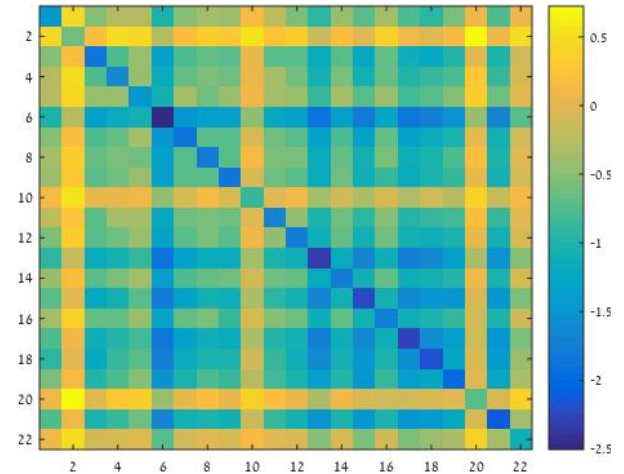
Assignment 6

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1A



1B

The pairing of the neurons showcases greater redundancy relative to the stimuli in the experiment.

There are 193 redundant pairs (83.55%) versus only 38 pairs of neurons (16.45%) who show synergy.

Moreover, the total sum of all the neuron pairs is negative (-142.1384) which indicates redundancy.

2B

According to the graphs, the noise correlation between neurons 14 and 17 varies between the 8 directions.

While directions 1-3 have a positive correlation, the others' correlations are negative. (the highest correlation is in the third direction which is close to 1).

2D

Signal correlation refers to the level of similarity between the mean responses of two neurons to some set of stimuli.

For example, given two direction tuning curves for the neurons, one can simply compute Pearson's correlation coefficient between the values for the two curves.

If the neurons prefer the same direction, then large responses from one neuron imply large responses from the other, and the r-value will be positive, near one.

If the neurons prefer opposite directions, then the r-value may be close to negative one.

Woo-Hoo!